

China's Growing Space Debris Problem

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Image Source: European Space Agency

A Rising Threat In LEO

- China is abandoning used launcher rocket stages in low Earth orbit (LEO) at an unprecedented and increasing pace
- From January 2021 to January 2025, China abandoned 51 spent rocket bodies above 650 km in altitude
 - Accounts for a massive 86% of the global total in that timeframe
- The mass of abandoned rocket bodies from China has more than tripled in recent years
 - Significant rise from 98,000 kg to 305,000 kg
 - This represents 98% of the global increase in abandoned mass
- By contrast, in the same period.....
 - The United States left only four rocket bodies
 - Russia left only one rocket body



Image Source: Deep in Space – The Science Board

LEO Explosions: A Unique Issue



Image Source: Getty Images

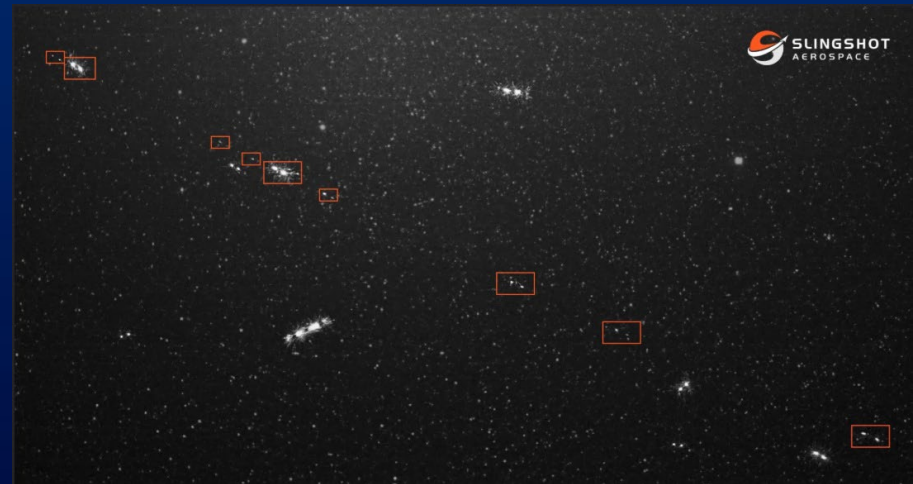
- The derelicts are often second-stage or kick rockets used to push payloads into high orbits before they begin their operations
 - Left behind with residual fuel that can cause explosions
- Three Chinese rocket bodies have already exploded in the last four years
 - Two CZ-6A rockets and one Zhuque-2
 - Residual debris will linger for decades to centuries
- Chinese rockets are generally larger than those of other nations
 - Significantly more dangerous fragmentation debris when they explode or collide in orbit
- These payloads primarily include satellites for China's expanding Guowang and Qianfan constellations
 - Designed as domestic alternatives to the Starlink broadband network
 - Only 200 of a planned 15,000 satellites (!) launched so far
 - So.....many more of these to come

Some Context and Examples

- China is not the worst offender in terms of number of objects or total mass
- However, net change over the last five years is more than 30x the rest of the world
- Many of these rocket bodies are at 800 km to 850 km altitude
 - US Space Defense Agency's Proliferated Warfighter Space Architecture (PWSA) will operate at ~1000 km
- Long March 6A Rocket
 - August 2024 launch
 - Explosion resulted in over 1000 pieces of trackable debris at ~810 km altitude
- Zhuque 2E Rocket
 - June 2026 breakup
 - 100 pieces of debris
 - Direct threat to ISS and Starlink constellation

R/Bs by Country	Above 650 km in LEO					
	01Jan21		01Jan26		Net Change	
	#	kg	#	kg	#	kg
China	41	98k	96	305k	+51	+207k
Russia	437	786k	438	787k	+1	+~1k
US	167	67k	171	67k	+4	+<1k
ROW	38	37k	41	41k	+3	+4k
Total	683	988k	746	1,200k	+59	+212k

Table Source: *BreakingDefense.com*



Long March 6A Rocket Debris

Source: *Slingshot Aerospace*

International Law and Potential Solutions

- International best-practice guidelines mandate a number of mitigation measures to reduce the risks of on-orbit explosions
 - Signed by more than 60 nations including China
 - Verification of compliance is difficult
- Potential mitigator – Left Over Fuel:
 - Use left-over fuel to maneuver spent rocket stages to low altitudes
 - Stages they fall back to Earth within 25 years naturally, or....
 - Preferably, in a controlled manner
- Potential mitigator – Orbit Injection:
 - Jettison spent launcher stages in a lower orbit to ensure 25-year deorbit
 - Rely on electric or other propulsion more slowly reach operational altitude
- PRC claims to already follow international best practices (June 11, 2026 Statement to UN Committee on Peaceful Uses of Outer Space)



Image Source: Center for Ethics and Rule of Law

References

- [1] Hitchens, T., "China Dumping More Rocket Bodies in Space, Endangering Low Earth Orbit Satellites: Report," Breaking Defense, June 25, 2026, <https://breakingdefense.com/2026/06/china-dumping-more-rocket-bodies-in-space-endangering-low-earth-orbit-satellites-report/>, Accessed July 3, 2026
- [2] Lemman, J., "Here's Why It Was So Hard to Track That Chinese Rocket That Fell to Earth," Popular Mechanics, May 27, 2021, <https://www.popularmechanics.com/space/rockets/a36344131/chinese-rocket-crash-tracking-space-objects>, Accessed July 9, 2026
- [3] Clark, S., "US Military Tracks More Than 300 Pieces of Debris from Chinese Launch," Ars Technica, Aug. 9, 2024, <https://arstechnica.com/space/2024/08/us-military-tracks-more-than-300-pieces-of-debris-from-chinese-launch>, Accessed July 10, 2026